Talk 1: Group-completion K-theory

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1 Monoids and groups

Proposition 1.1. $i\colon \mathrm{Grp} \hookrightarrow \mathrm{Mon}$ *has a left adjoint.*

Proof. Take $M \mapsto M/R_M$ (quotient by group relations). □

Remark 1.2. In fact, the same works for E_1- and E_∞-monoids.

Definition 1.3. Let $\mathcal{C} \in \mathrm{Cat}_\infty$ have finite products. A *cartesian monoid* in $\mathcal{C}$ is a functor $X\colon \Delta^{op} \to \mathcal{C}$ such that

1. $X_0 \simeq \mathrm{pt}$,
2. $X_n \simeq \underbrace{X_1 \times \cdots \times X_1}_{n \text{ factors}}$, via the Segal maps induced by the inclusions $[1] \to [n]$, $0 \mapsto i-1$, $1 \mapsto i$.

We write $\mathrm{Mon}(\mathcal{C}) \subseteq \mathrm{Fun}(\Delta^{op}, \mathcal{C})$ for the full subcategory of cartesian monoids. The multiplication is $X_1 \times X_1 \simeq X_2 \xrightarrow{d_1} X_1$, and the unit is $\mathrm{pt} \simeq X_0 \xrightarrow{s_0} X_1$; the higher X_n encode associativity and all its higher coherences.

Remark 1.4. In Topologie IV, symmetric monoidal ∞-categories were defined as coCartesian fibrations $\mathcal{C}^\otimes \to \mathrm{Fin}_*$ satisfying a Segal condition [2, Def. 5.18], i.e. by straightening [2, Thm. 5.32], functors $\mathrm{Fin}_* \to \mathrm{Cat}_\infty$ with $\mathcal{C}_{\langle n \rangle} \simeq \mathcal{C}_{\langle 1 \rangle}^n$. Replacing Fin_* by Δ^{op}, an object of $\mathrm{Mon}(\mathrm{Cat}_\infty)$ is a *monoidal* ∞-category, i.e. one without a symmetry.

Example 1.5. For $\mathcal{C} = \mathrm{Set}$ we get $\mathrm{Mon}(\mathrm{Set}) \simeq \mathrm{Mon}$: a monoid M corresponds to its nerve $[n] \mapsto M^n$, i.e. the nerve of the one-object category BM; the face maps multiply adjacent entries and the degeneracies insert units.

Definition 1.6. An E_1-*monoid* is an object of $\mathrm{Mon}(\mathrm{An})$.

Definition 1.7. $X \in \mathrm{Mon}(\mathcal{C})$ is called a *group* if the shear map

$$X_1 \times X_1 \longrightarrow X_1 \times X_1, \qquad (f, g) \longmapsto (f, f \cdot g)$$

is an equivalence. We write $\mathrm{Grp}(\mathcal{C}) \subseteq \mathrm{Mon}(\mathcal{C})$.

Here $f \cdot g$ denotes the multiplication $X_1 \times X_1 \simeq X_2 \xrightarrow{d_1} X_1$. For $\mathcal{C} = \mathrm{Set}$ this is the usual notion: the preimage of $(f, 1)$ is (f, f^{-1}).

Remark 1.8. A monoid $X \in \mathrm{Mon}(\mathrm{An})$ is a group if and only if the ordinary monoid $\pi_0 X_1$ is a group.

Theorem 1.9. *The inclusion* $\mathrm{Grp}(\mathrm{An}) \hookrightarrow \mathrm{Mon}(\mathrm{An})$ *has a left adjoint, the group completion* $M \mapsto M^{gp}$ *[3, Thm. 2.1.7]. We will see below that* $M^{gp} \simeq \Omega BM$.

The next theorem is used below to construct the nerve adjunction between simplicial anima and ∞-categories.

Theorem 1.10. *Let $\mathcal{D}$ be cocomplete and $\mathcal{C}$ small, and consider the restriction along the Yoneda embedding $y_{\mathcal{C}}\colon \mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{op}, \mathrm{An})$:*

$$\mathrm{Fun}(\mathrm{Fun}(\mathcal{C}^{op}, \mathrm{An}), \mathcal{D}) \xrightarrow{y_{\mathcal{C}}^*} \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

1. *A functor F on the left-hand side preserves colimits if and only if it has a right adjoint.*
2. *$y_{\mathcal{C}}^*$ restricts to an equivalence $\mathrm{Fun}^L(\mathrm{Fun}(\mathcal{C}^{op}, \mathrm{An}), \mathcal{D}) \xrightarrow{\simeq} \mathrm{Fun}(\mathcal{C}, \mathcal{D})$, where Fun^L denotes the colimit-preserving functors.*

In a slogan: $\mathrm{Fun}(\mathcal{C}^{op}, \mathrm{An})$ is the *free cocompletion* of $\mathcal{C}$. Colimit-preserving functors out of it are the same as arbitrary functors out of $\mathcal{C}$.

Remark 1.11. In the equivalence above, $\mathrm{Lan}_y F \longleftrightarrow F$, with

$$(\mathrm{Lan}_y F)(X) \simeq \mathrm{colim}_{(c,\ y(c)\to X)} F(c), \qquad \text{right adjoint } d \longmapsto \mathrm{Map}_{\mathcal{D}}(F(-), d)$$

[3, Thm. 1.5.18].

2 ∞-categories as simplicial anima

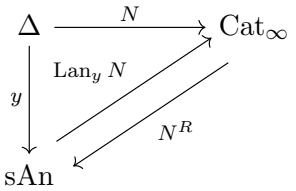
Definition 2.1. $N\colon \Delta \to \mathrm{Cat}_\infty$, $[n] \mapsto N([n])$ (the nerve of the poset $[n]$, regarded as an ∞-category), gives

$$\mathrm{Fun}(\Delta^{op}, \mathrm{An}) \longrightarrow \mathrm{Cat}_\infty$$

by left Kan extension (Theorem 1.10 with $\mathcal{C} = \Delta$, $\mathcal{D} = \mathrm{Cat}_\infty$). And the right adjoint is given by

$$N^R\colon \mathrm{Cat}_\infty \longrightarrow \mathrm{Fun}(\Delta^{op}, \mathrm{An})$$

the *Rezk nerve*. We write $\mathrm{sAn} := \mathrm{Fun}(\Delta^{op}, \mathrm{An})$ for simplicial anima. The whole picture:



with $\mathrm{Lan}_y N \circ y \simeq N$ and $\mathrm{Lan}_y N \dashv N^R$.

Remark 2.2. $N^R(\mathcal{C})_n \simeq \mathrm{Fun}(\Delta^n, \mathcal{C})^{\simeq} =: \mathrm{Map}_{\mathrm{Cat}_\infty}(\Delta^n, \mathcal{C})$, by the formula for the right adjoint in Remark 1.11.

Theorem 2.3 (Rezk, Joyal–Tierney)**.** *N^R is fully faithful with an essential image consisting of the complete Segal anima, i.e. $X \in \mathrm{sAn}$ is* Segal *if*

$$X_n \simeq X_1 \times_{X_0} \cdots \times_{X_0} X_1,$$

via the Segal maps, and complete *as in Definition 2.5 below.*

Remark 2.4. In particular $N^R\colon \mathrm{Cat}_\infty \xrightarrow{\simeq} \mathrm{CSAn}$ is an equivalence.

Definition 2.5. A Segal anima X is *complete* if $s_0 \colon X_0 \to X_1^{\times}$ is an equivalence, where $X_1^{\times} \subseteq X_1$ consists of the components of the invertible arrows. We write $\mathrm{CSAn} \subseteq \mathrm{sAn}$ for the complete Segal anima.

Remark 2.6 (isomorphisms are equivalent to identities). Let X be a Segal anima. For $x, y \in X_0$ there are two notions of "x and y are the same":

1. a path $x \simeq y$ in the anima X_0;
2. an *invertible arrow* $f \in X_1$ with $d_1 f = x$, $d_0 f = y$, i.e. one having a left and a right inverse for the composition $X_1 \times_{X_0} X_1 \simeq X_2 \xrightarrow{d_1} X_1$.

Identities are invertible, so s_0, $x \mapsto \mathrm{id}_x$, lands in $X_1^{\times}$. Completeness, $s_0 \colon X_0 \xrightarrow{\simeq} X_1^{\times}$, says that every invertible arrow is equivalent (by a path in X_1) to an identity, in an essentially unique way.

This is forced in the image of N^R. If $X \simeq N^R(\mathcal{C})$, then $X_0 \simeq \mathrm{Map}([0], \mathcal{C}) = \mathcal{C}^{\simeq}$ and $X_1 \simeq \mathrm{Map}(\Delta^1, \mathcal{C}) = \mathrm{Fun}(\Delta^1, \mathcal{C})^{\simeq}$, and s_0 sends x to id_x. An arrow $f \in \mathrm{Fun}(\Delta^1, \mathcal{C})^{\simeq}$ is equivalent to some id_x if and only if f is an equivalence in $\mathcal{C}$ (an equivalence $\mathrm{id}_x \simeq f$ in $\mathrm{Fun}(\Delta^1, \mathcal{C})$ is a commuting square whose vertical maps are equivalences). So $X_1^{\times}$ is exactly the anima of equivalences in $\mathcal{C}$, and $s_0 \colon X_0 \to X_1^{\times}$ hits every component. That it is an equivalence is the statement $\mathrm{Map}([0], \mathcal{C}) \simeq \mathrm{Map}(J, \mathcal{C})$, where J is the walking equivalence, which is clear as J is contractible.

Example 2.7. E_1-monoids are Segal: since $X_0 \simeq \mathrm{pt}$, the fibre products are just products and everything follows from the Segal condition.

Remark 2.8. E_1-monoids are usually not complete: $X_0 \simeq \mathrm{pt}$, while $X_1^{\times}$ consists of the invertible components of X_1. For example, a group $G \in \mathrm{Grp}(\mathrm{An})$ is complete only if $G \simeq \mathrm{pt}$. Turning a Segal anima into a complete one is done by the completion functor of Theorem 2.9 below.

Theorem 2.9. $\mathrm{CSAn} \subseteq \mathrm{sAn}$. *There is a functor*

$$\mathrm{sAn} \xrightarrow{\mathrm{Lan}_y N} \mathrm{Cat}_\infty \xrightarrow{N^R} \mathrm{CSAn} \subseteq \mathrm{sAn}.$$

It is a left adjoint to the inclusion $\mathrm{CSAn} \hookrightarrow \mathrm{sAn}$. *It is called* completion, $\mathrm{comp} := N^R \circ \mathrm{Lan}_y N$.

Proof. Let $X \in \mathrm{sAn}$ and $Y \in \mathrm{CSAn}$. By Theorem 2.3, $Y \simeq N^R(\mathcal{D})$ for some $\mathcal{D} \in \mathrm{Cat}_\infty$, and N^R is fully faithful. Hence

$$\mathrm{Map}_{\mathrm{CSAn}}\big(N^R \mathrm{Lan}_y N(X), Y\big) \simeq \mathrm{Map}_{\mathrm{Cat}_\infty}\big(\mathrm{Lan}_y N(X), \mathcal{D}\big) \simeq \mathrm{Map}_{\mathrm{sAn}}\big(X, N^R(\mathcal{D})\big) \simeq \mathrm{Map}_{\mathrm{sAn}}(X, Y),$$

the first equivalence by full faithfulness and the second by the adjunction $\mathrm{Lan}_y N \dashv N^R$; all are natural in X and Y [1, Lemma I.67]. □

Remark 2.10. For a Segal anima X, $\mathrm{comp}(X)_0 = (\mathrm{Lan}_y N(X))^{\simeq}$ is X_0 with the invertible arrows of X_1 added as paths [3, Prop. 1.8.4].

3 Monoids and groups as pointed ∞-categories

Theorem 3.1. $\mathrm{Mon}(\mathrm{An}) \xrightarrow{\mathrm{comp}} \big(\mathrm{CSAn}_{*/}\big)_{\geq 1}$ *is fully faithful with an essential image* $\big(\mathrm{CSAn}_{*/}\big)_{\geq 1}$. *The functor sends M to its completion* $\mathrm{comp}(M)$, *pointed by* $\mathrm{pt} \simeq M_0 \to \mathrm{comp}(M)_0$, *and* $(-)_{\geq 1}$ *means that X_0 is connected [3, Cor. 2.1.2].*

Since $\mathrm{CSAn} \simeq \mathrm{Cat}_\infty$, we get

$$\mathrm{Mon}(\mathrm{An}) \xrightarrow{\simeq} \big((\mathrm{Cat}_\infty)_{*/}\big)_{\geq 1},$$

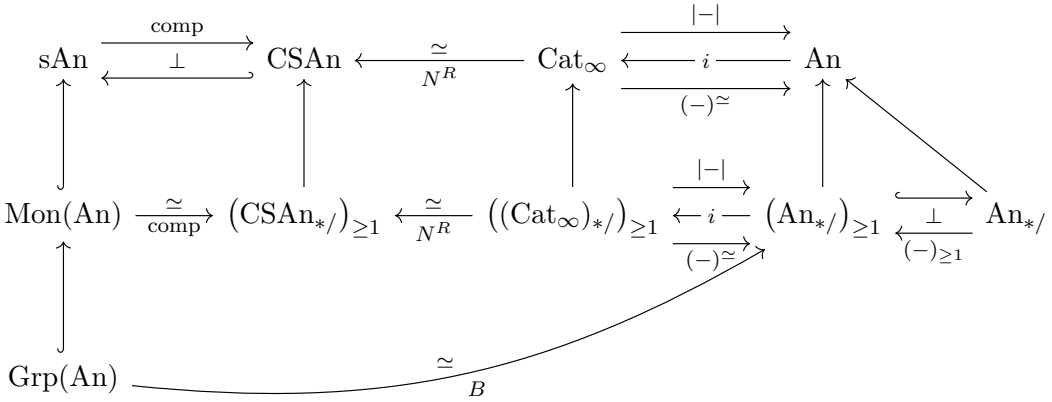
the pointed ∞-categories $(\mathcal{C}, x)$ in which every object is equivalent to x. Slogan: an E_1-monoid is the same as an ∞-category with one object (up to equivalence).

Remark 3.2. $\big((\mathrm{Cat}_\infty)_{*/}\big)_{\geq 1} \longrightarrow \mathrm{Mon}(\mathrm{An})$, $(\mathcal{C}, x) \longmapsto \mathrm{Map}_{\mathcal{C}}(x, x)$, with multiplication given by composition. As a simplicial anima, $\mathrm{Map}_{\mathcal{C}}(x, x)$ is $[n] \mapsto N^R(\mathcal{C})_n \times_{(\mathcal{C}^\simeq)^{n+1}} \{(x, \dots, x)\}$, the n-simplices all of whose vertices are x.

Definition 3.3. For $M \in \mathrm{Mon}(\mathrm{An})$ let $BM := \mathrm{colim}_{\Delta^{op}} M$ (the geometric realisation of the simplicial anima M), pointed by $\mathrm{pt} \simeq M_0 \to BM$; this gives a functor $B\colon \mathrm{Mon}(\mathrm{An}) \to \mathrm{An}_{*/}$ [3, Cor. 2.4.1]. For a discrete group G, $BG = K(G, 1)$.

Theorem 3.4. $\mathrm{Grp}(\mathrm{An}) \xrightarrow{\simeq} \big(\mathrm{An}_{*/}\big)_{\geq 1}$, $G \mapsto BG$, *with inverse* $(X, x) \mapsto \Omega_x X = \mathrm{Map}_X(x, x)$ *(recognition principle, Stasheff; [3, Cor. 2.4.1]).*

Remark 3.5 (B is a delooping). The whole picture:



The upper composite of the top row is $\mathrm{colim}_{\Delta^{op}}$, the lower one is the constant functor. The upper composite of the middle row is B, the one through i is Ω, so $B \dashv \Omega$, and on groups $G \simeq \Omega BG$. Under the equivalence of the middle row, the inclusion $\mathrm{Grp}(\mathrm{An}) \subseteq \mathrm{Mon}(\mathrm{An})$ corresponds to i.

Claim 3.6. *There is a right adjoint* $(-)^\simeq$ *of* $i\colon \mathrm{An} \hookrightarrow \mathrm{Cat}_\infty$*, the core.*

Remark 3.7. The above picture explains an equivalence, namely Theorem 3.4:

$$\begin{gathered}\text{an } \infty\text{-category is an anima iff all morphisms are invertible} \\ \iff \text{each monoid } \mathrm{Map}_{\mathcal{C}}(x, x) \text{ is a group}\end{gathered}$$

So the equivalence of Theorem 3.1 restricts to $\mathrm{Grp}(\mathrm{An}) \simeq (\mathrm{An}_{*/})_{\geq 1}$ [3, Cor. 2.4.1].

4 E_∞-monoids (commutative monoids)

Definition 4.1. Γ^{op} is the 1-category with objects the finite sets and morphisms $X \to Y$ the partially defined maps, i.e. a subset $A \subseteq X$ together with a map $A \to Y$; composition is composition where defined [3, Def. 2.5.1]. Sending the undefined elements to an added base point, $X \mapsto X_+$, gives $\Gamma^{op} \simeq \mathrm{Fin}_*$ [2, Def. 5.11]. We write $\langle n \rangle = \{1, \dots, n\}$.

Construction 4.2. There is a map $\Delta^{op} \to \Gamma^{op}$, $[n] \mapsto \langle n \rangle$ (called Cut in [3, Def. 2.5.1]). Think of $j \in \langle n \rangle$ as the arrow $j-1 \to j$ of $[n]$. A map $\alpha\colon [m] \to [n]$ in Δ sends the arrow $i-1 \to i$ to the path from $\alpha(i-1)$ to $\alpha(i)$, and goes to the partial map

$$\langle n \rangle \longrightarrow \langle m \rangle, \qquad j \longmapsto i \ \text{ if } \alpha(i-1) < j \leq \alpha(i), \quad \text{undefined if there is no such } i.$$

Example 4.3. For the three maps $d^0, d^1, d^2\colon [1] \to [2]$: the map d^1 ($0 \mapsto 0$, $1 \mapsto 2$) sends the only arrow of $[1]$ to the path through both arrows of $[2]$,

$$\begin{array}{lccc} [1]\colon & 0 & \xrightarrow{\quad 1 \quad} & 1 \\ & \downarrow & & \downarrow \\ [2]\colon & 0 & \xrightarrow[1]{} 1 \xrightarrow[2]{} & 2 \end{array}$$

so it goes to $\langle 2 \rangle \to \langle 1 \rangle$, $1, 2 \mapsto 1$. For a monoid M this is the multiplication $M_1 \times M_1 \simeq M_2 \to M_1$, $(m_1, m_2) \mapsto m_1 m_2$. Similarly d^2 ($0 \mapsto 0$, $1 \mapsto 1$) goes to $1 \mapsto 1$, 2 undefined, the projection $(m_1, m_2) \mapsto m_1$, and d^0 to the projection $(m_1, m_2) \mapsto m_2$. But Γ^{op} has more maps than come from Δ^{op}, e.g. the flip $\langle 2 \rangle \to \langle 2 \rangle$, $1 \mapsto 2$, $2 \mapsto 1$; this will encode commutativity.

Definition 4.4. $X\colon \Gamma^{op} \to \mathrm{An}$ is called an *E_∞-monoid* if $\Delta^{op} \to \Gamma^{op} \xrightarrow{X} \mathrm{An}$ is an E_1-monoid.

Remark 4.5 (relation to symmetric monoidal ∞-categories). The same definition with Cat_∞ in place of An gives $\mathrm{CMon}(\mathrm{Cat}_\infty)$, and by straightening a functor $\Gamma^{op} \simeq \mathrm{Fin}_* \to \mathrm{Cat}_\infty$ with the Segal condition is just a symmetric monoidal ∞-category [2, Def. 5.18]. E_∞-monoids are precisely those landing in $\mathrm{An} \subseteq \mathrm{Cat}_\infty$, i.e. symmetric monoidal ∞-categories whose underlying ∞-category is an anima.

Notation 4.6. E_∞-monoids $= \mathrm{CMon}(\mathrm{An})$, E_∞-groups $= \mathrm{CGrp}(\mathrm{An})$, those whose underlying E_1-monoid is a group.

Theorem 4.7.

$$\mathrm{CGrp}(\mathrm{An}) \underset{\hookrightarrow}{\overset{\Omega B}{\dashleftarrow}} \mathrm{CMon}(\mathrm{An})$$

$\mathrm{CGrp}(\mathrm{An}) \subseteq \mathrm{CMon}(\mathrm{An})$ *has a left adjoint ΩB, the group completion $M \mapsto M^{gp}$ [3, Prop. 2.5.13, Cor. 2.5.15].*

Construction 4.8. Two ways to see M^{gp} for $M \in \mathrm{CMon}(\mathrm{An})$:

- $M^{gp} \simeq \Omega BM$: take the group completion ΩBM of the underlying E_1-monoid (Theorem 1.9). It is again an E_∞-monoid, hence an E_∞-group, and it is M^{gp}; i.e. forgetting from E_∞ to E_1 commutes with group completion [3, Cor. 2.4.2, Cor. 2.5.15].
- M^{gp} is the pushout in $\mathrm{CMon}(\mathrm{An})$ [3, Prop. 2.5.13]

$$\begin{array}{ccc} M & \xrightarrow{\Delta} & M \times M \\ \downarrow & \ulcorner & \downarrow \\ \mathrm{pt} & \longrightarrow & M^{gp} \end{array}$$

  A pair (a, b) stands for "$a - b$", and the diagonal pairs (m, m) are set to 0.

Remark 4.9. $\pi_0(M^{gp}) \cong \pi_0(M)^{gp}$ is the Grothendieck group of $\pi_0(M)$, since $\pi_0 \Omega BM = \pi_1 BM$. For example $\mathbb{N}^{gp} \simeq \mathbb{Z}$, as $B\mathbb{N} \simeq S^1$.

Example 4.10. $\mathrm{Proj}_R \in \mathrm{CMon}(\mathrm{An})$: the groupoid of finitely generated projective R-modules and their isomorphisms, considered as an anima, with $\oplus$ and 0. It is an E_∞-monoid by Remark 4.5, as $(\mathrm{Proj}(R), \oplus, 0)$ is a symmetric monoidal category.

Definition 4.11. Let R be a ring. Then

$$K(R) := \mathrm{Proj}_R^{gp} \in \mathrm{CGrp}(\mathrm{An}), \qquad K_i(R) := \pi_i\big(K(R)_1, 0\big)$$

Here $K(R)_1 = K(R)(\langle 1\rangle)$ is the underlying anima [3, Def. 2.3.1].

Example 4.12. Consider the composite Q:

$$\begin{array}{ccccc} \Delta^{op} & \xrightarrow{\mathrm{Sing}_*(\mathbb{C})} & \mathrm{Ring} & \longrightarrow & \mathrm{CMon}(\mathrm{An}) \\ [n] & \longmapsto & C^0(|\Delta^n|, \mathbb{C}) & \longmapsto & \mathrm{Proj}_{C^0(|\Delta^n|,\mathbb{C})} \end{array}$$

where $C^0(|\Delta^n|, \mathbb{C})$ is the ring of continuous functions and $R \mapsto \mathrm{Proj}_R := \mathrm{Proj}(R)^{\simeq}_{f.g.}$ as in Example 4.10 (functorial via $P \mapsto P \otimes_R S$ for $R \to S$). Put

$$\mathrm{Vect}_{\mathbb{C}} := \mathrm{colim}_{\Delta^{op}} Q \in \mathrm{CMon}(\mathrm{An}).$$

Theorem 4.13. $\mathrm{Vect}_{\mathbb{C}} \simeq \coprod_{n\geq 0} BU(n)$ *in* $\mathrm{CMon}(\mathrm{An})$*, where the right hand side is an* E_∞*-monoid via block sum* $U(n) \times U(m) \to U(n+m)$*.*

Sketch. Write $R_k = C^0(|\Delta^k|, \mathbb{C})$. As $|\Delta^k|$ is contractible, finitely generated projective R_k-modules are free (Swan), so the groupoid Proj_{R_k} has one object R_k^n for each n, with automorphism group $GL_n(R_k)$. As an anima, a one-object groupoid with automorphisms G is $BG = \mathrm{colim}_{[j]\in\Delta^{op}} G^j$ (Section 3), so $Q_k \simeq \coprod_n B\,GL_n(R_k)$. Now commute the two colimits:

$$\mathrm{Vect}_{\mathbb{C}} \simeq \coprod_n \mathrm{colim}_{[k]}\, \mathrm{colim}_{[j]}\, GL_n(R_k)^j \simeq \coprod_n \mathrm{colim}_{[j]} \big(\mathrm{colim}_{[k]}\, GL_n(R_k)\big)^j \simeq \coprod_n B\big(\mathrm{colim}_{[k]}\, GL_n(R_k)\big),$$

using that Δ^{op} is sifted, i.e. $\mathrm{colim}_{\Delta^{op}}$ commutes with finite products in An (this is also why the colimit in $\mathrm{CMon}(\mathrm{An})$ is computed in An [3, Obs. 2.5.9]). Finally $GL_n(R_k)$ is the set of continuous maps $|\Delta^k| \to GL_n(\mathbb{C})$, so $[k] \mapsto GL_n(R_k)$ is the singular complex of the topological group $GL_n(\mathbb{C})$ and $\mathrm{colim}_{[k]}\, GL_n(R_k) \simeq GL_n(\mathbb{C})^{top} \simeq U(n)$ (Gram–Schmidt). Hence $\mathrm{Vect}_{\mathbb{C}} \simeq \coprod_n BU(n)$, with $\oplus$ becoming block sum. □

Remark 4.14. This is the example of topological K-theory: $\mathrm{Vect}_{\mathbb{C}}$ is the topologised version of $\mathrm{Proj}_{\mathbb{C}}$, and $\mathrm{Vect}_{\mathbb{C}}^{gp} = K^{top}(\mathbb{C}) \simeq \mathbb{Z} \times BU \in \mathrm{CGrp}(\mathrm{An})$ [3, Def. 2.3.3].

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